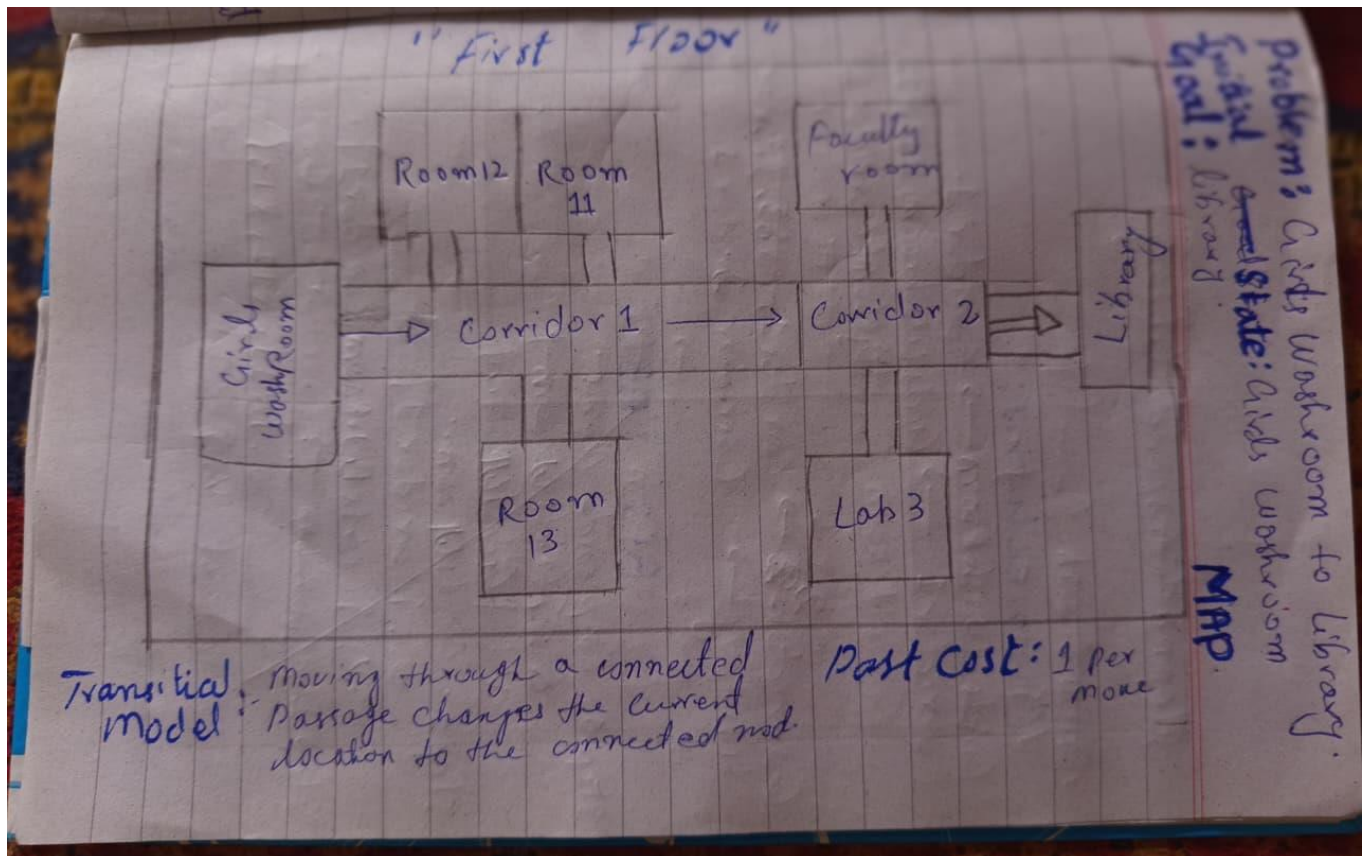




Connections:

Girls Wash Room $\rightarrow$ Corridor 1.
 Corridor 1 $\rightarrow$ Girls Washroom, R11, R12, R13, Faculty Room, Corridor 2.
 Corridor 2 $\rightarrow$ Corridor 1, Lab 3, Library.
 Library $\rightarrow$ Corridor 2.

Step 0: Expand None.

$Q: \{ \text{Girls W. Room} \}$

$V: \{ \text{Girls W. Room} \}$

Step 1:

Expand Girls W. Room.

$Q: \{ \text{Corridor 1} \}$

$V: \{ \text{Girls W. Room, Corridor 1} \}$

Step 2: Expand Corridor 1

$Q: \{ R11, R12, R13, \text{Faculty Room, Corridor 2} \}$

$V: \{ \text{G.W.R., Corridor 1, R12, R11, R13, Faculty R., Corridor 2} \}$

Step 3: Expand R12.

$Q: \{ R11, R13, \text{Faculty R., Corridor 2} \}$

$V: \{ \text{No new node} \}$

Step 4: Expand Room 11.

$Q: \{ R13, \text{Faculty Room, Corridor 2} \}$

$V: \{ \text{No new node} \}$

Step 5: Expand R13.

$Q: \{ \text{Faculty Room, Corridor 2} \}$

$V: \{ \text{No new node} \}$

Step 6: Expand Corridor 2

$Q: \{ \text{Library} \}$

$V: \{ \text{add Library} \}$

Step 8. Expand library.

Q: { Goal found? }
V: { - }

Parent Table

Node	Parent
Girls WRoom	→ None
Corridor 1	→ None Girls WRoom
R-12	→ Girls WRoom Corridor 1
R-11	→ Corridor 1
R-13	→ "
Faculty Room	→ "
Corridor 2	→ "
Library	→ Corridor 6.

Shortest Path:

Girls WRoom → Corridor 1 → Corridor 2



Path Cost = 3 moves

Library